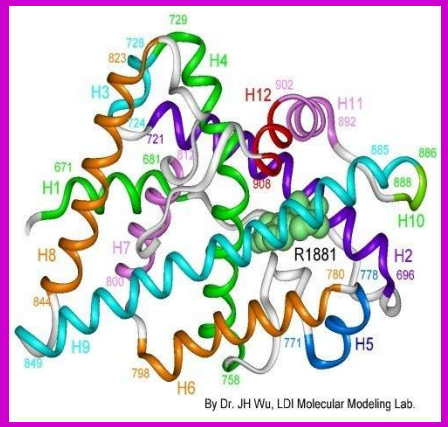
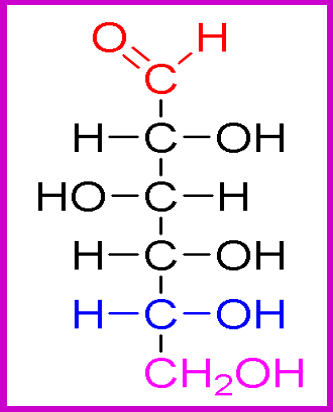
BIOCHEMISTRY IN DENTISTRY





Introduction

- Biochemistry is the science concerned with the chemical basis of life.
- It is also the science concerned with the chemical constituents of living cells and with the reactions and processes they undergo.
- It is the application of the principles and methods of chemistry to the field of biology and physiology.
- It is the language of biology basic to the understanding of the different phenomena both in the biological and medical sciences.
- Biochemistry encompasses large areas of cell biology, molecular biology and molecular genetics.

What is Biochemistry?

- **Biochemistry** is a branch of medical science that seeks to describe the structure, organization and functions of living matter in molecular terms.
- **It is the chemistry of life**. It is divided into 3 principal areas:
- 1. Structural chemistry
- 2. Metabolism
- 3. Chemistry of molecular genetics

Roots of Biochemistry

- **Karl Scheele** – Swedish founder of biochemistry. He studied the chemical composition of matter in mid 1700.
- **Schleiden & Schwann** – formulated the cell theory in 1840.
- **Walter Flemming** – discovered chromosomes in
- 1875
- **Carl Newberg** – a German scientist who coined the word biochemistry
- **Hans Kreb** – Proposed the Kreb cycle of the TCA in 1937.
- **Embden & Mayerhoff** – described the glycolytic pathway in 1925.
- **James Watson & Francis Crick** – described the double helical structure of DNA in 1953

Roots of Biochemistry

- **Edward & Hans Buchner** – found that extracts from yeasts could bring about fermentation of sugar into ethanol in 1897
- **Paul Boyer and J. Walker** – discovered the "rotary engine" that generated ATP in 1997.
- **Danish J. Skou-** studied the "pump" that drives sodium and potassium across membranes
- **Stanley Prusiner** – discovered the organism that caused "mad cow disease."
- **Ruska, et.al**. – discovered the electron microscope and provided a whole new level of insight into cellular structure.

A Knowledge of Biochemistry is essential to All Life Processes

- The biochemistry of nucleic acids lies at the heart of Genetics; application of genetic engineering and cloning
- Physiology overlaps with biochemistry almost completely
- Immunology employs numerous biochemical techniques/approaches
- Pharmacology and pharmacy rest on sound knowledge of biochemistry in the creation of “designer drugs” or drug architecture
- Invention of new drugs in Pharmacy, Medicine, Agriculture and other fields
- Used in Environmental Science
- Importance in Biology (zoology & botany) and in microbiology for many scientists
- Biochemical approaches are employed in Pathology
- Poisons act on biochemical reactions and this is the subject matter in toxicology.

Biochemistry in Relation to Dentistry

The aims, attitudes and techniques of biochemistry are as relevant to dentistry as to medicine or to any aspect of biology.

3. To understand the true nature of dental disease. All diseases have a biochemical basis.

5. To give dental patients the necessary or appropriate dietary advice to prevent dental disease.

3. Special relevance to dentists are areas of blood coagulation and effects of drugs and other injected substances on tissue and cells.

Relevance of Biochemistry to Dentistry

4. Understanding the physicochemical process of resorption and deposition of bone minerals and its matrix is essential to orthodontics
5. As for the future, methods to prevent or cure tooth decay are likely to involve a biochemical approach, like caries vaccine.
6. The role of flouride is now well established and its role to remineralize a carious lesion or chemically modifying a tooth, the enamel surface and its bacterial population offer scope for further investigation

Methods of Determining Biomolecular Structures

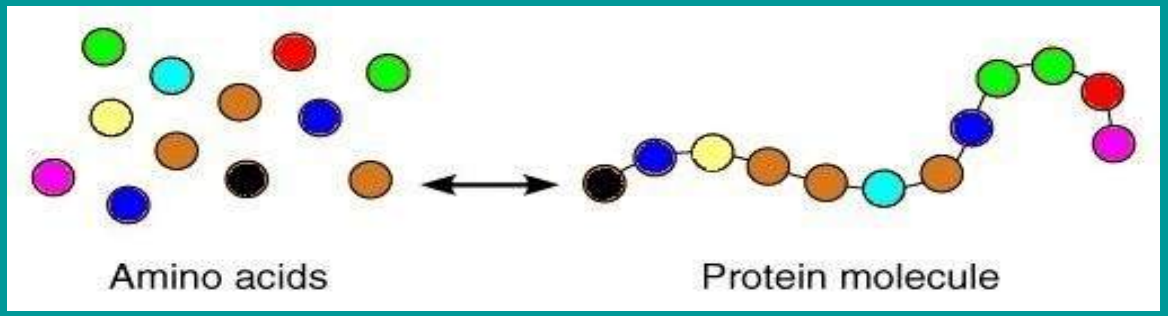
- Elemental analysis
- UV, visible, infrared, and NMR spectroscopy
- Mass Spectroscopy
- X-ray Crystallography
- Specific sequencing methods (e.g., for proteins and nucleic acids)
- Use of battery of enzymes of known specificity to degrade the biomolecule under study
- Use of acid or alkaline hydrolysis to degrade the biomolecule under study

Differences between Living and Non-Living Things

1. They are complicated and highly organized.
2. Each part of a living organism appears to have a specific purpose of function
3. They are able to extract energy from the environment
4. They are capable of reproducing themselves through generations
5. They exhibit common properties of living matter

What are Biomolecules?

- Biomolecules are molecules found in living matter.



- Two broad types: Small molecules and macromolecules
- Importance of Macromolecules:

a) Essential structures for the basis of life

b) Control and regulate these processes

c) Responsible for energy exchanges,

What are the Primordial Biomolecules?

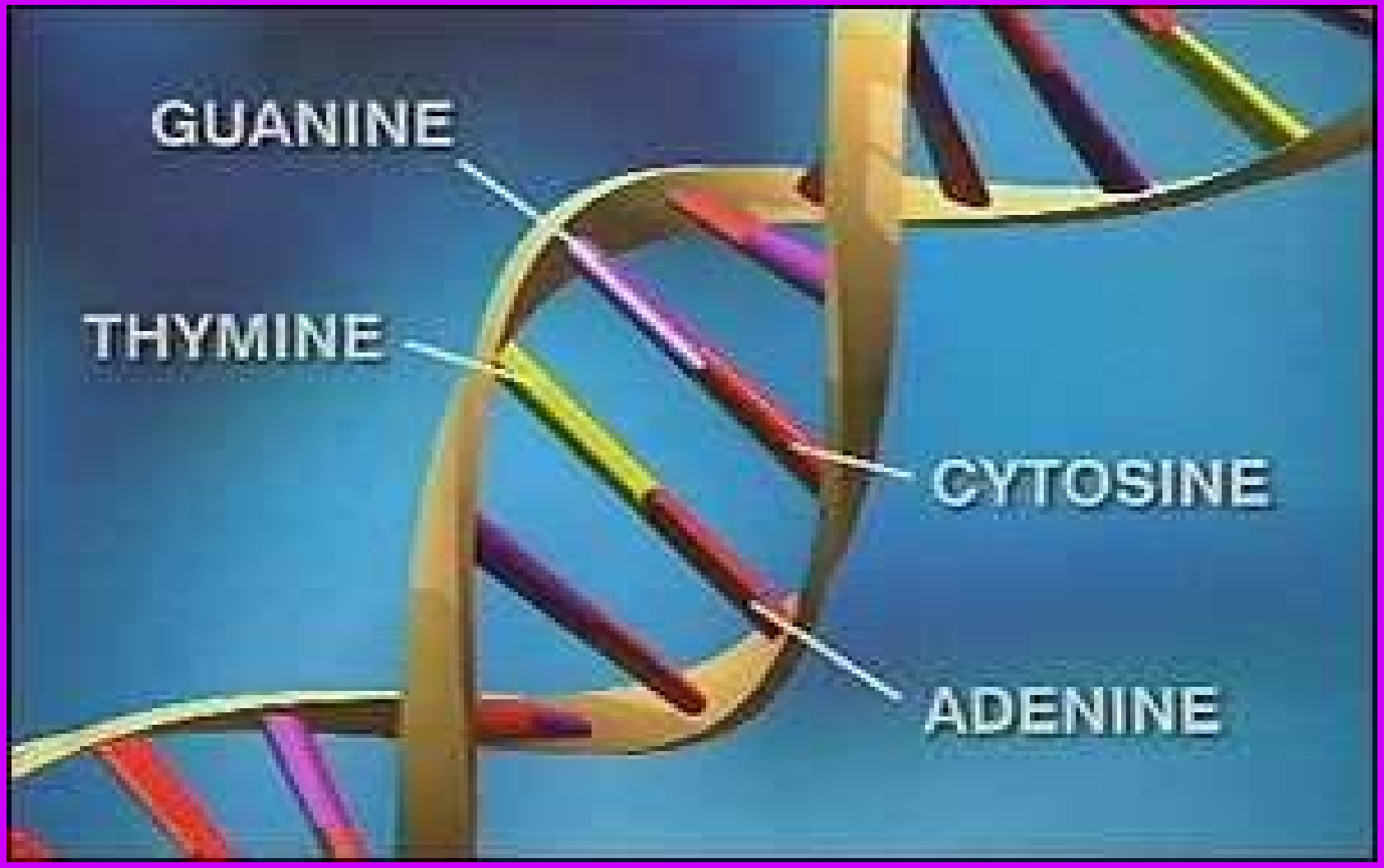
1. Amino Acids – glycine, alanine, serine
2. Nitrogenous bases – pyrimidines, purines
3. Sugars – glucose, galactose, mannose
4. Sugar alcohol - glycerol
5. Nitrogenous alcohol - choline
6. Fatty acids – palmitic acid, linoleic acid, linolenic acid, arachidonic acid



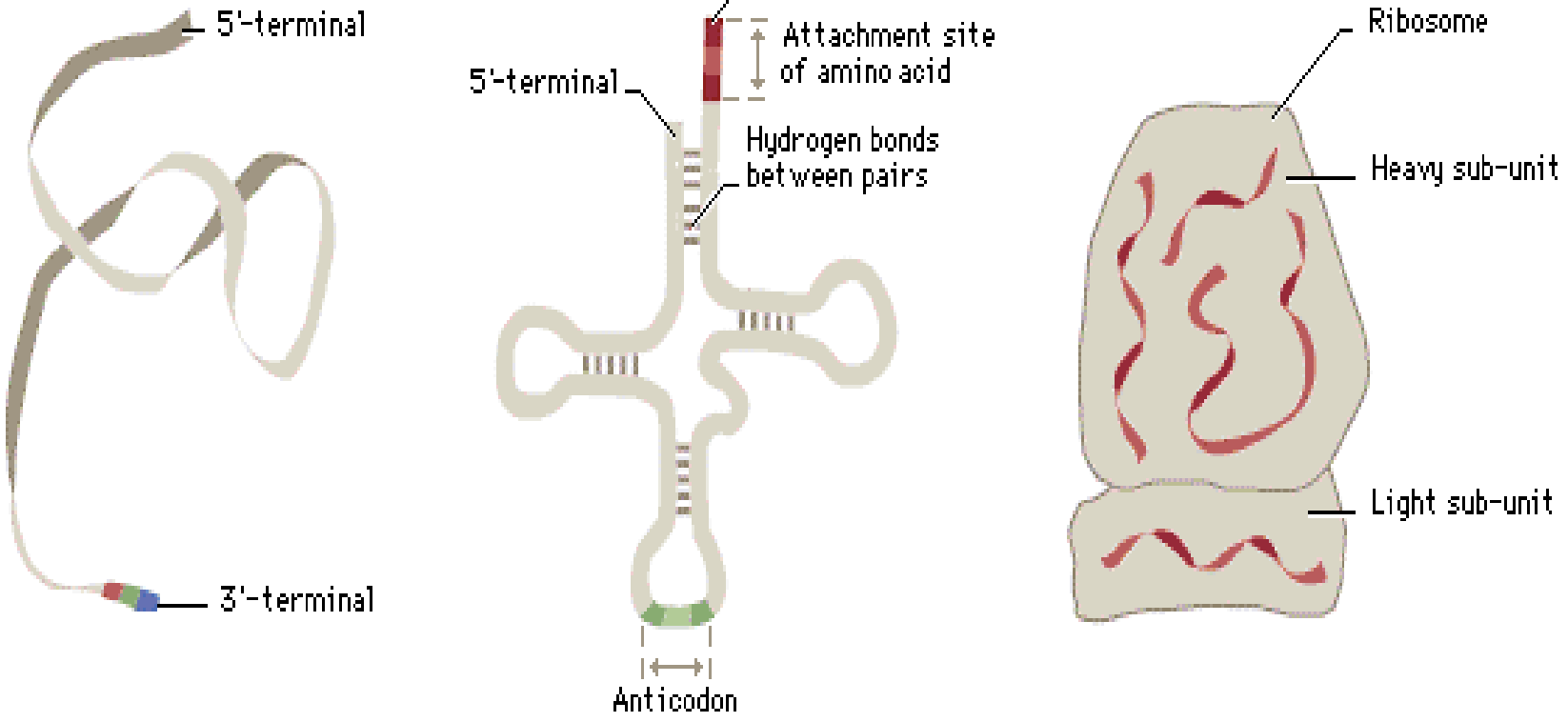
Amino acids are organic compounds made in the body which combine to form proteins

A.D.A.M.

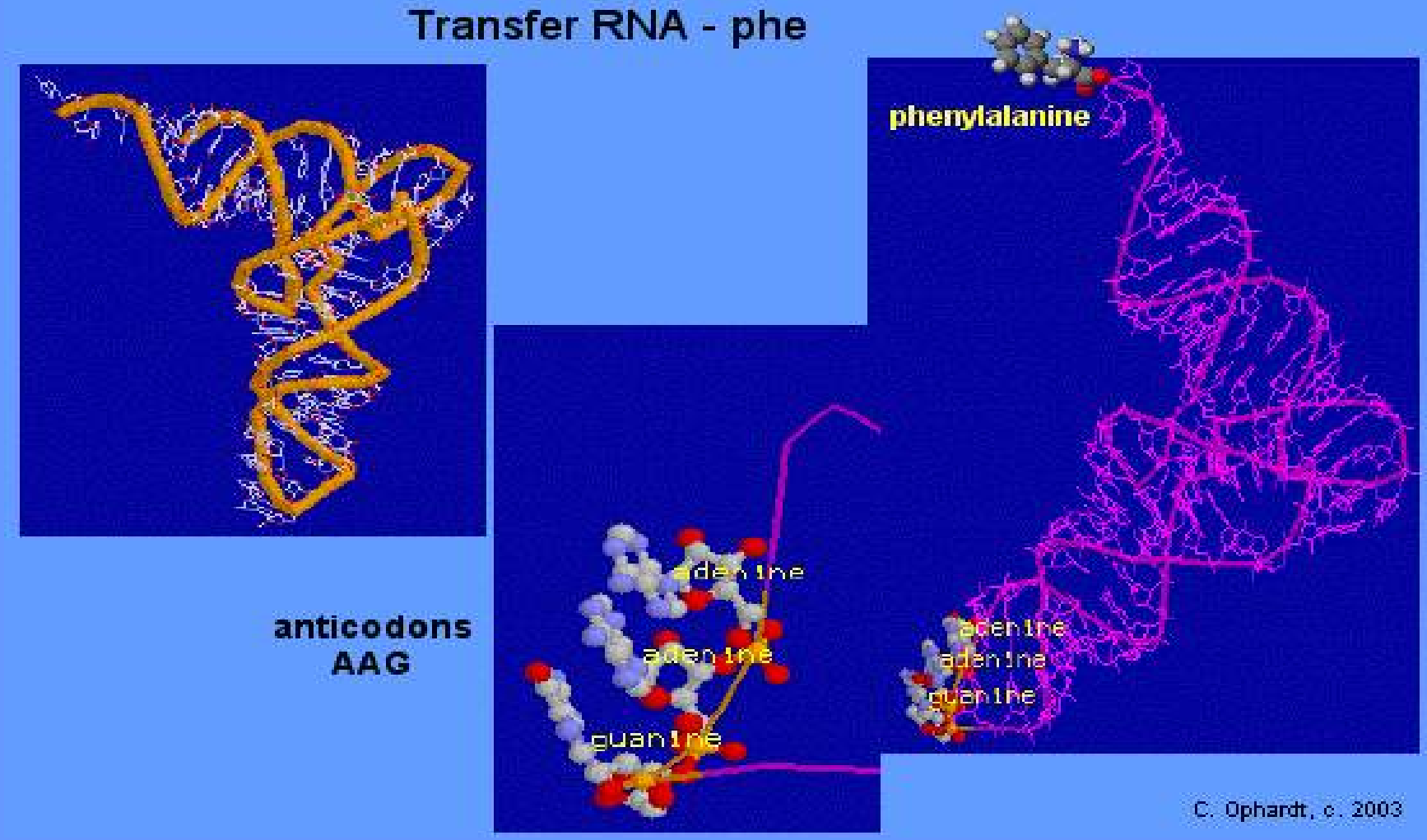
DNA



RNA


5'-terminal
3'-terminal
3'-terminal
Attachment site
of amino acid
5'-terminal
Hydrogen bonds
between pairs
Anticodon
Ribosome
Heavy sub-unit
Light sub-unit
Messenger RNA
Transfer RNA
Ribosomal RNA

tRNA


Transfer RNA - phe
phenylalanine
anticodons
AAG
adenine
adenine
guanine
adenine
adenine
guanine
C. Ophardt, c. 2003

Chemical Composition of Living Matter

- **Water** – 70-90% (free and bound water)
- **Solids** – 10-30%
- **Inorganic substances** – 1% (Na, K, Ca, Mg, NH_4, Cl^-, SO_4, PO_4^{-3}, CO_3^{-2}, etc.

- **Traces of Fe, I_2, Cu, Mn, Co, Zn** are also present in combination with organic radicals
- **Rest**- organic substances

Water

- This is the major component of the cell and is often referred to as an inert space filter in a living organism.
- It is a strong dipole and has a high dielectric constant.
- It is highly reactive with unusual properties different physically and chemically from other common liquids.
- Water and its ionization products H^+ and OH^- are important factors in determining the structure and biological properties of proteins, nucleic acids, lipids, and other cell components.

Properties of Water of Biological Importance

- It is a universal solvent
- It is an ideal biologic agent or medium for the ionization of substances and therefore hastens chemical reactions
- It has a high specific heat, that is, it takes up more heat to raise its temperature through 1°C, thus allowing the body to store heat effectively without greatly raising its temperature.
- It possesses a high latent heat of evaporation
- It has the capacity to conduct heat readily

Water as an ideal biologic agent

- Water is a dipole, a molecule with chemical charge distributed asymmetrically about its structure.
- Hydrogen bonding enables water to dissolve many organic biomolecules that contain functional groups which can participate in hydrogen bonding.
- Hydrogen bonds account for the surface tension, viscosity, liquid state at room temperature, and solvent power of water.
- Compounds that contain O, N or S can serve as hydrogen bond donors or acceptors.

pH

- pH is the negative log of the hydrogen ion concentration.
- pH = -log(H^+)
- Low H values correspond to high concentration of H^+ and high pH values correspond to low concentrations of H^+.
- Acids are proton donors and bases are proton acceptors
- Strong acids completely dissociate into anions and cations even in strongly acidic solutions.
- Strong bases are completely dissociated at high pH.
- Many biochemicals are weak acids.
- HCl and H_2SO_4 are strong acids
- KOH and NaOH are strong bases
- $Ca(OH)_2$ is a weak base

How to calculate for pH?

- What is the H of a solution whose hydrogen ion concentration is 3.2×10^{-4} mol/L?

$$pH = -\log (H^+)$$
$$= -\log (3.2 \times 10^{-4})$$
$$= -\log (3.2) - \log(10^{-4})$$
$$= -0.5 + 4.0$$
$$= 3.5$$

Solutions of Weak Acids and Their Salts Buffer Changes in pH

- Solutions of weak acids or bases and their conjugates exhibit buffering, the ability to resist a change in pH following addition of strong acid or base.
- Since many metabolic reactions are accompanied by the release or uptake of protons, most intracellular reactions are buffered.

Chemical Reactions Occurring in Living matter (In Vivo)

- Oxidation
- Reduction
- Hydrolysis
- Condensation
- Tautomerism

Oxidation

- Oxidation is the process wherein most of the energy liberated by living matter is derived from the oxidation of organic substances such as carbohydrates, fats and proteins
- Two kinds of oxidation: anaerobic oxidation and aerobic oxidation

Aerobic oxidation

- Aerobic oxidation takes place in the presence of free oxygen
- Example:

$2Zn + O_2 \longrightarrow 2ZnO$

Here the substance oxidized combines directly with oxygen

Anaerobic Oxidation

- In the absence of free oxygen, anaerobic oxidation occurs. In this case, the substance undergoes oxidation either by a loss of hydrogen, as in the oxidation of lactic acid to pyruvic acid.
- $CH_3CHOHCOOH \longrightarrow CH_3COCOOH$

  lactic acid

  pyruvic acid

Reduction

- Reduction is the reverse of oxidation. Hence, it may be brought about by either by loss of oxygen or by gain of hydrogen or electrons. It may be stated, therefore, that whenever oxidation occurs there is a simultaneous and corresponding reduction.
- All foods and organic substances have the property of taking up oxygen, hence they are reducing agents.

Hydrolysis

- **Hydrolysis is the union of a substance with one or more molecules of water, forming an unstable “substance-water-complex” which is subsequently fragmented.**
- **Through hydrolysis, large molecules are broken down into smaller and simpler forms.**

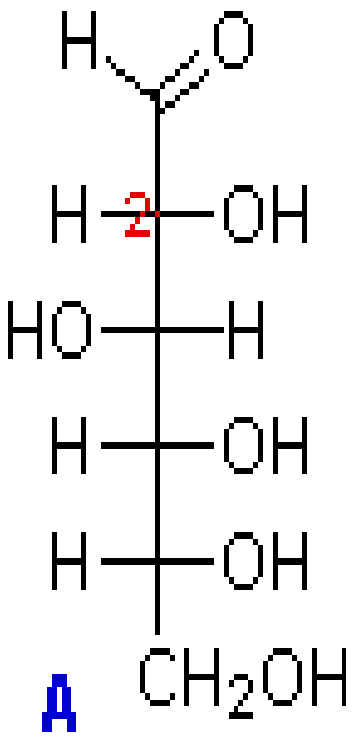
Condensation

- **Condensation is the reaction wherein simple fragments unite with one another to form a more complex compound.**
- **The synthesis of complex substances like glycogen and tissue protein is accomplished through this process.**

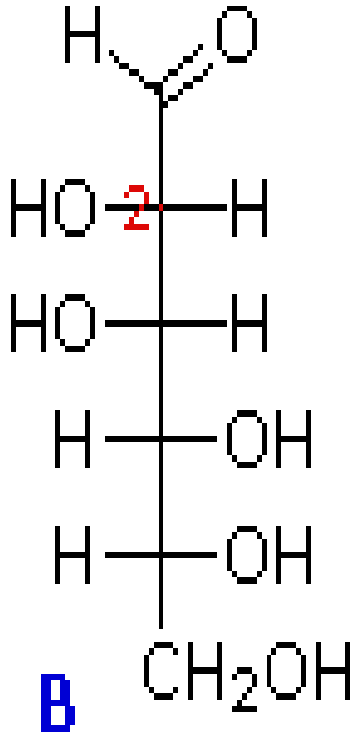
Tautomerism

- **Tautomerism or isomeric transformation is the intramolecular rearrangement of atoms within a molecule leading to the formation of a new substance having distinctive properties of its own.**
- **Example: transformation of glucose into galactose; galactose into mannose**

Glucose-Mannose



A



B

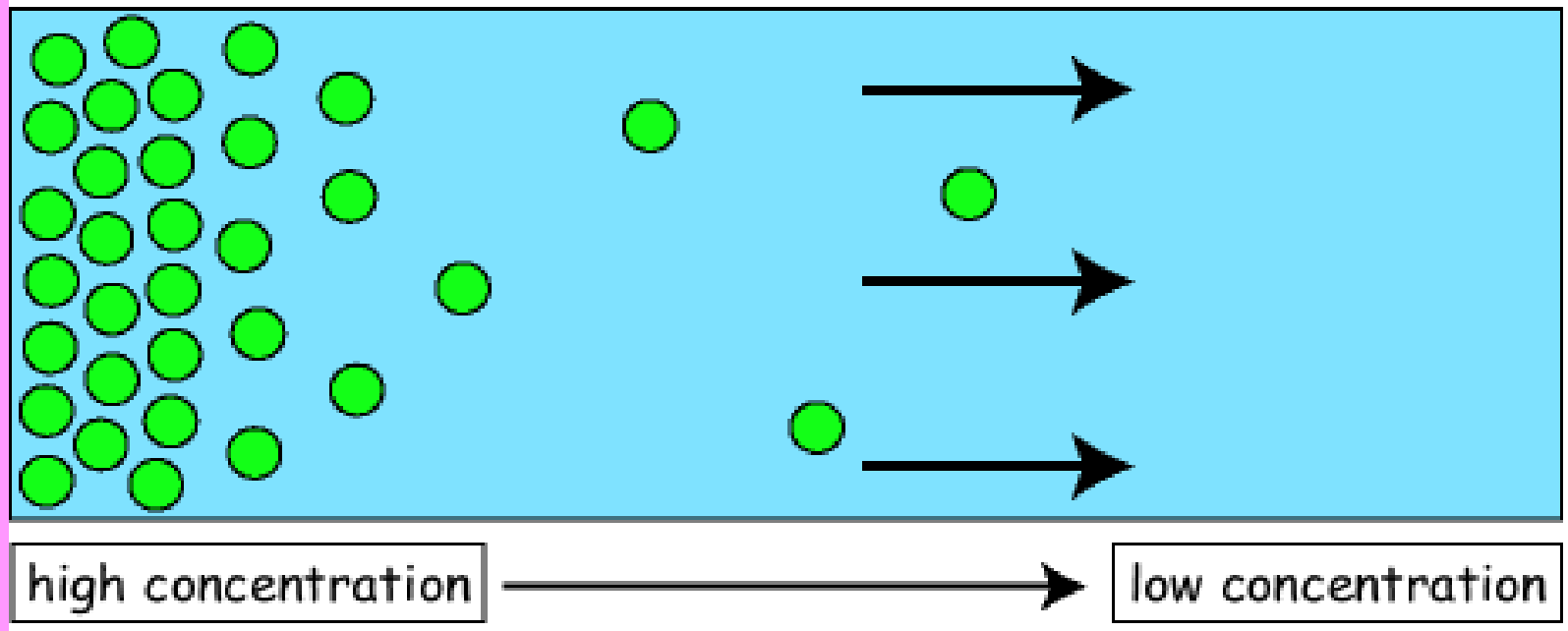
A pair of C-2 epimers.

One is glucose, the other is mannose.

Diffusion

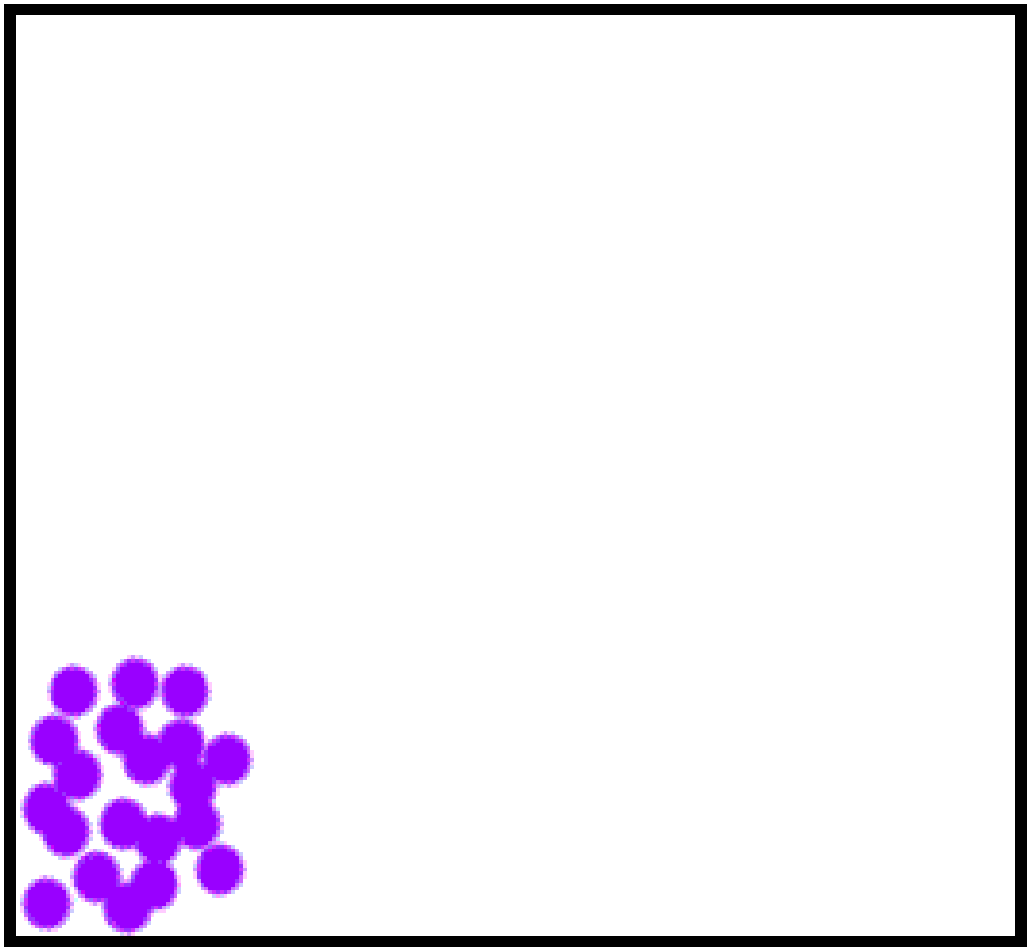
- **Diffusion is the interpenetration of molecules between two substances. This occurs whenever the solute distributes itself uniformly into the solvent.**
- **Diffusion is influenced by: size of molecules, temperature, moelecular weight**

Diffusion



- solute

Solute transport is from the left to the right; movement of the solutes is due to the concentration gradient (dC/dx).

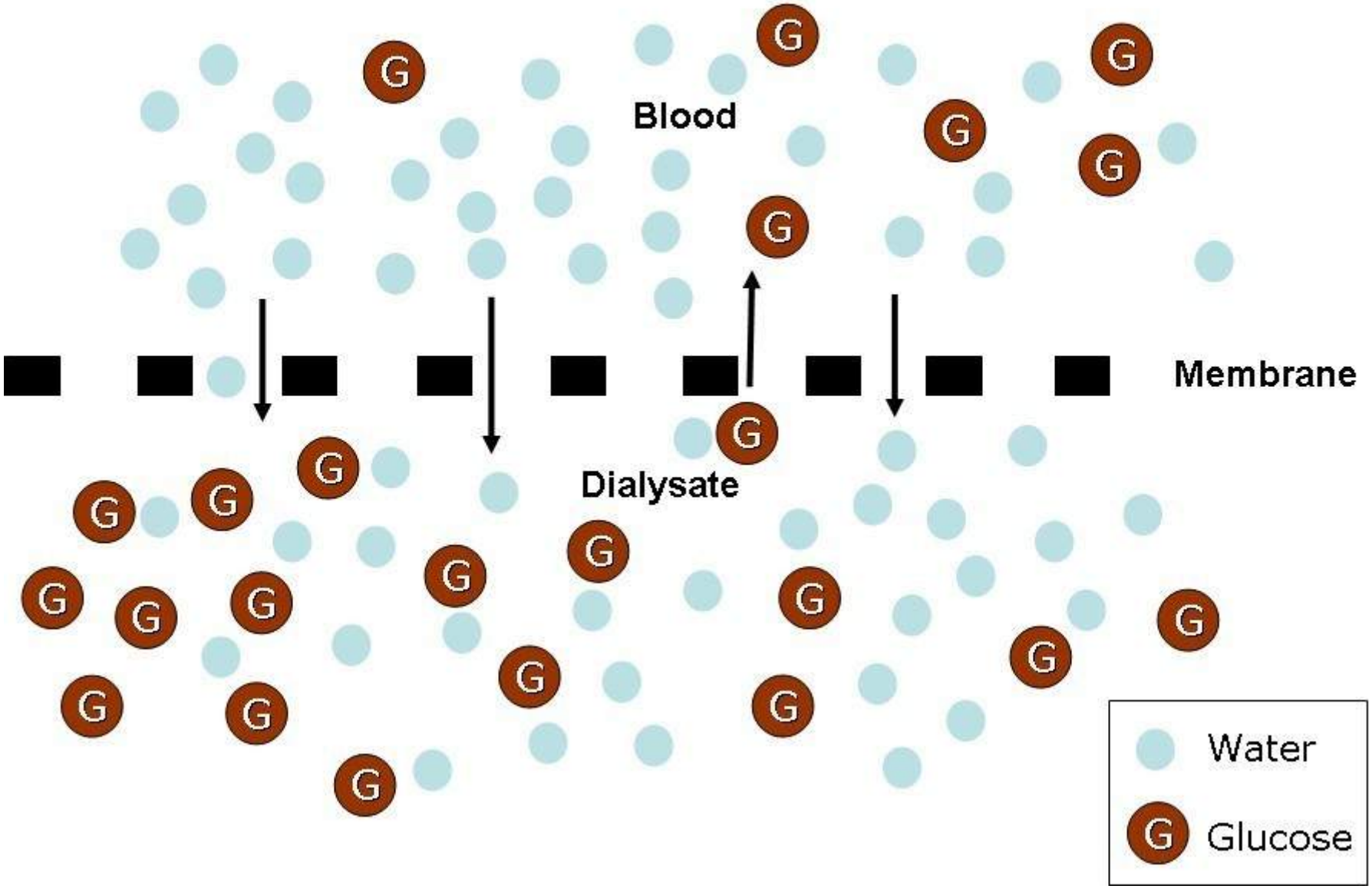


diffusion

Osmosis

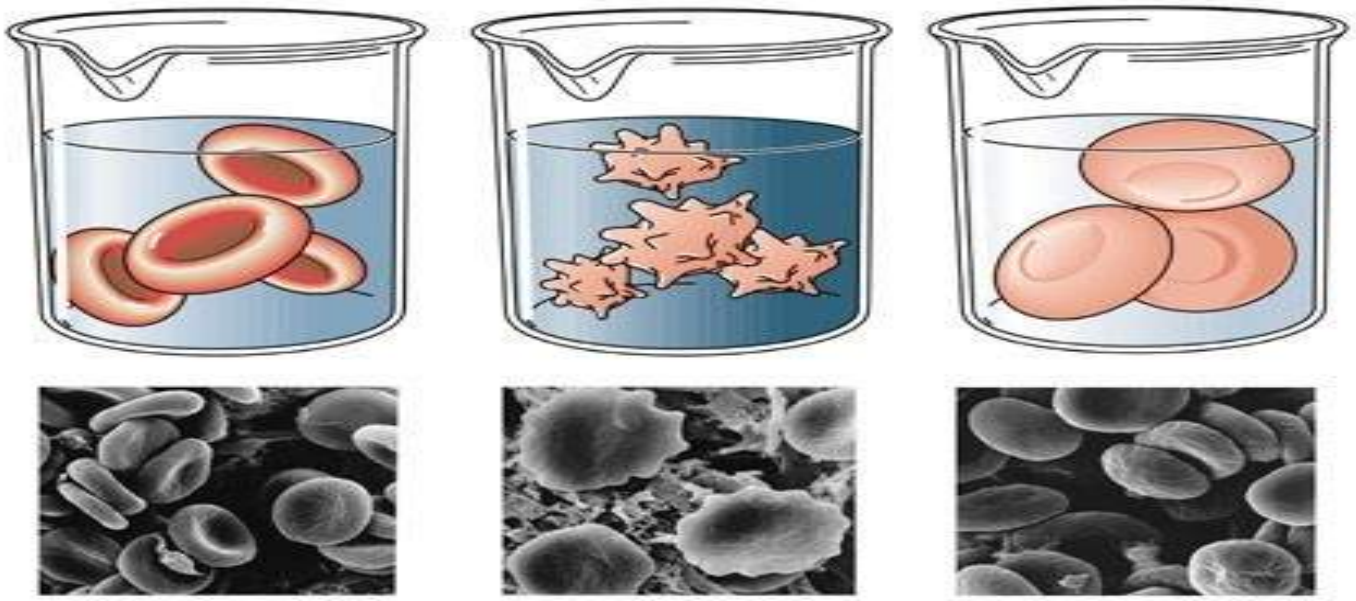
- Whenever two solutions of unequal concentration s are separated by a semi-permeable membrane, the fluid tends to flow from the side of low osmotic pressure to that of higher osmotic pressure until an osmotic equilibrium is reached.

Osmosis with glucose



Osmosis in RBC

Tobin/Dusheck, Asking About Life, 2/e
Figure 4.20

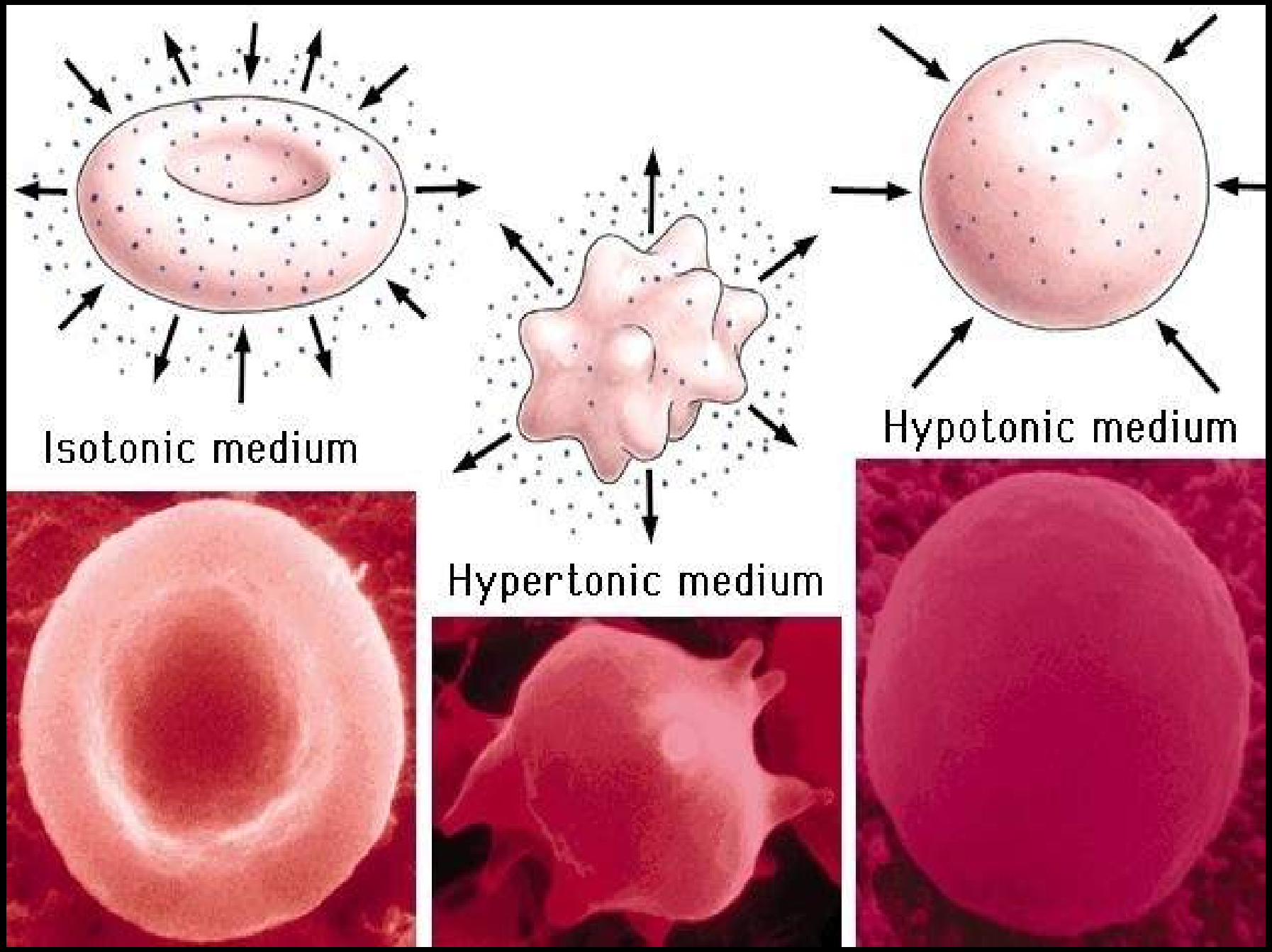


A.
Isotonic solution (equal concentration of ions in solution and cell)

B.
Hypertonic solution (higher concentration of ions in solution than in cell)

C.
Hypotonic solution (lower concentration of ions in solution than in cell)

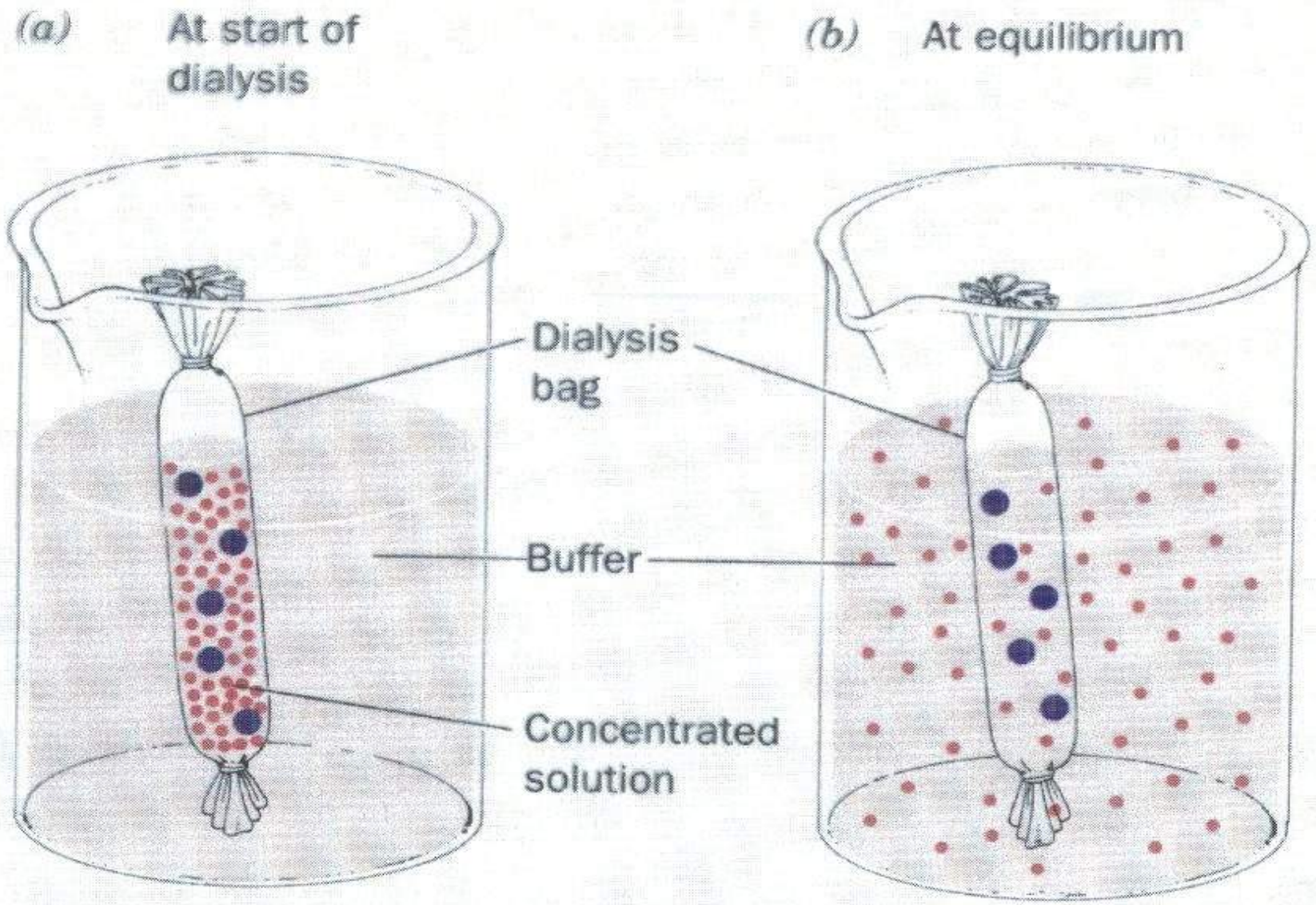
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Isotonic medium
Hypertonic medium
Hypotonic medium

Dialysis

- When two different solutions are separated by a membrane which allows the passage of the crystalloids but not the colloids, dialysis occurs.
- If a mixture of crystalloids and colloids is placed in a dialysing bag (collodion or parchment) and immersed in distilled water the crystalloids pass out while the colloids are left behind.
- This is utilized in the purification of colloids from crystalloid impurities or vice versa.

Dialysis


(a) At start of dialysis
(b) At equilibrium
Dialysis bag
Buffer
Concentrated solution

Surface Tension



Surface Tension

- **Molecules in the interior of a homogenous liquid are attracted on all directions by surrounding molecules so they move freely on all directions.**

- **The force by which the molecules are held is called the “surface tension.”**

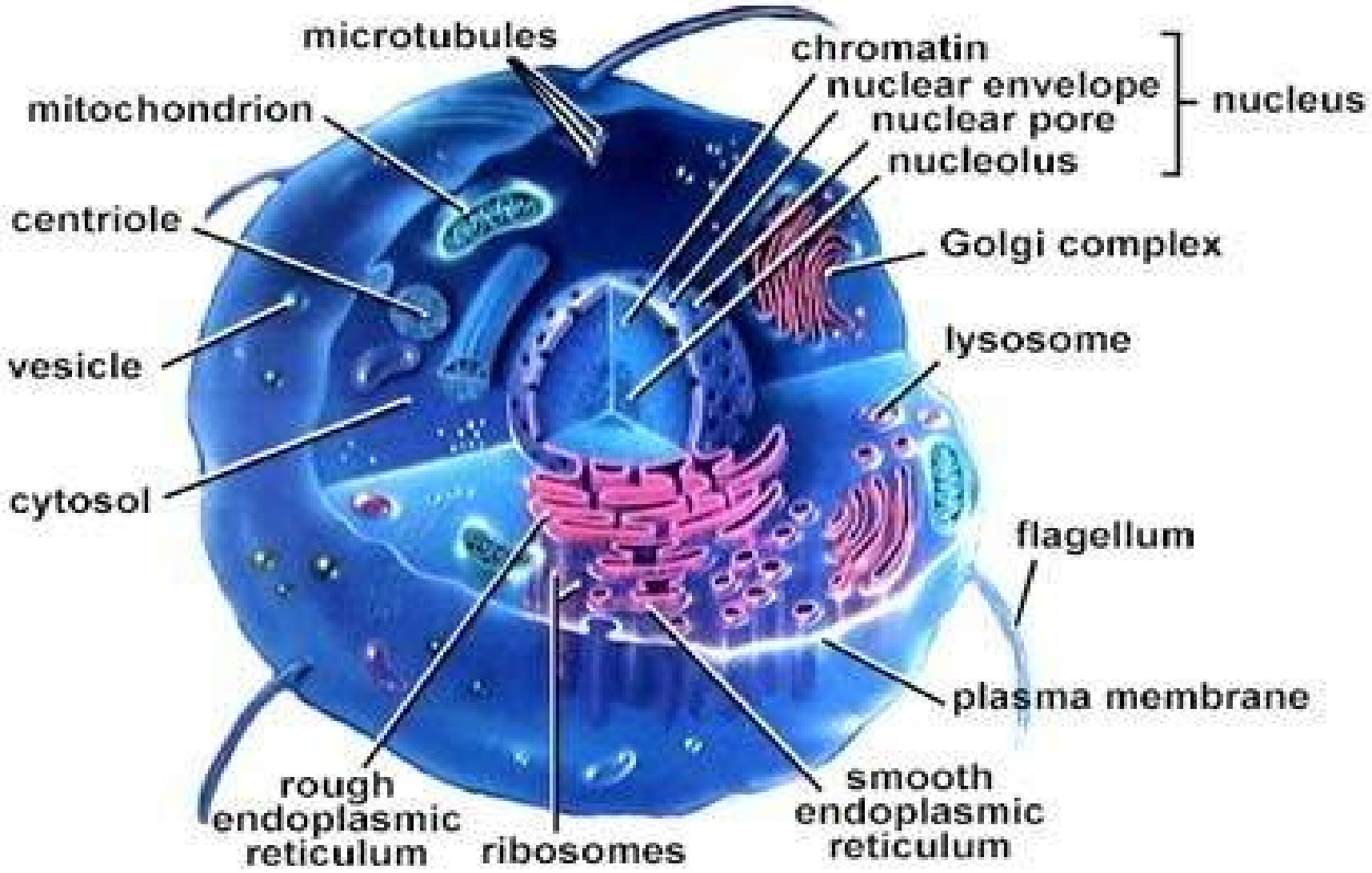
Surface Tension



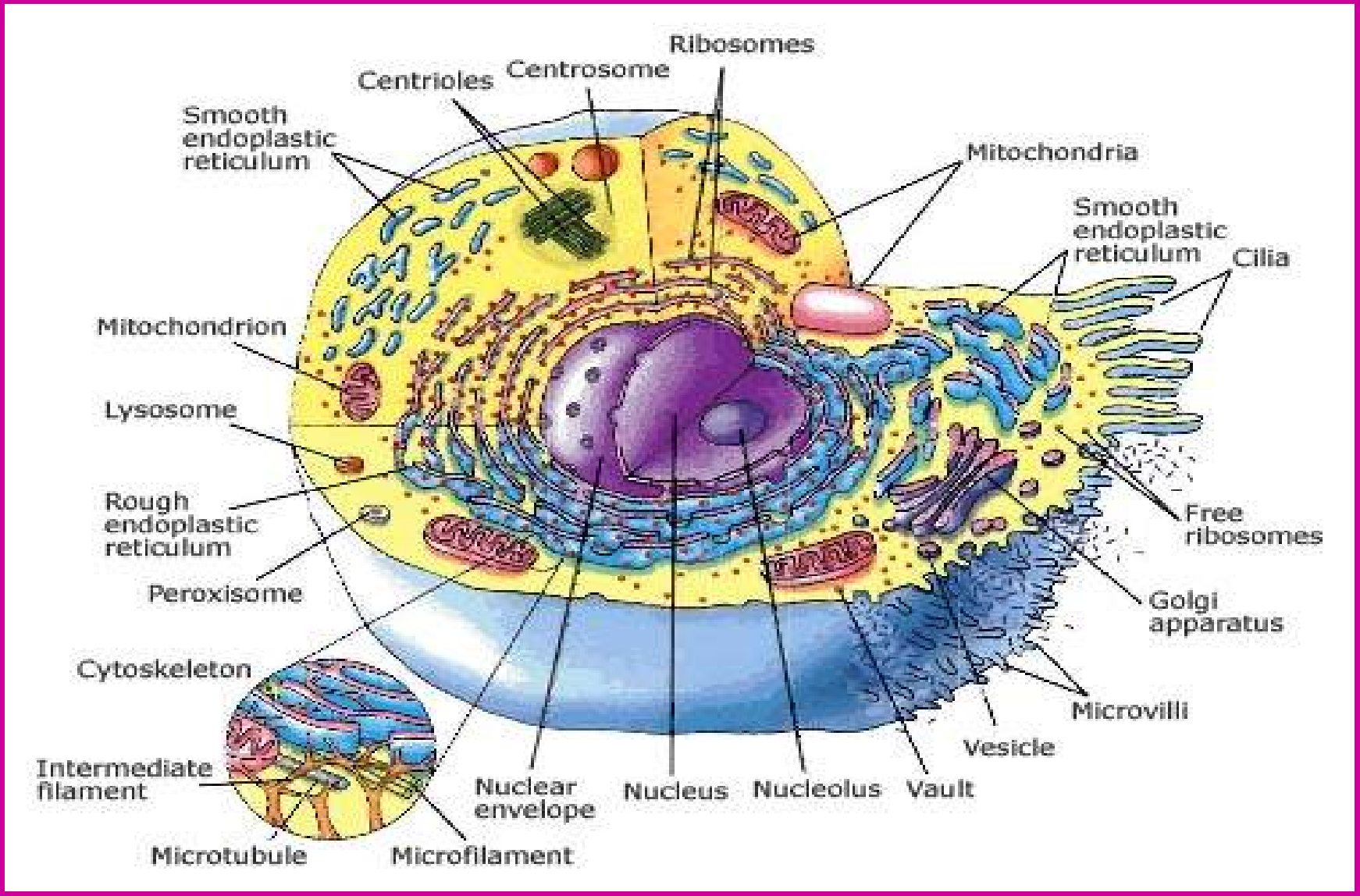
Hierarchy in the Molecular Organization of Cells

- **Precursors from the environment** (**CO_2, H_2O, ammonia, nitrogen**)
- **Metabolic intermediates** – (**puruvate, citrate, malate, glyceraldehyde-3-phosphate**)
- **Building blocks** (**nucleotides, amino acids, monosaccharides, fatty acids**)
- **Macromolecules** (**nucleic acids, proteins, polysaccharides, lipids**)
- **Supramolecular assemblies** (**ribosomes, enzyme complexes, contractile systems, microtubules**)
- **Organelles** (**nucleus, mitochondria, golgi complex, endoplasmic reticulum, lysosomes**)

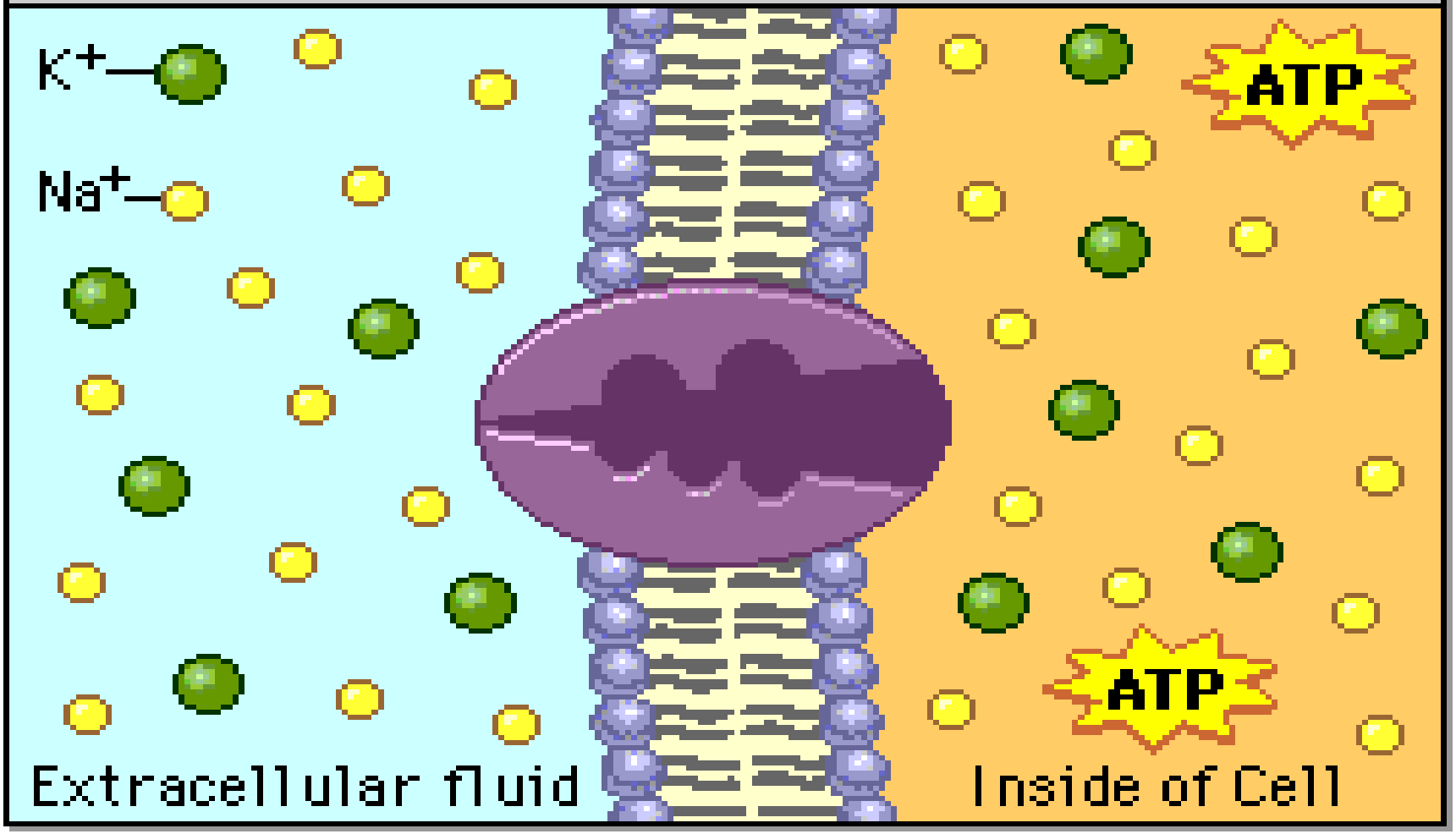
The Cell


microtubules
mitochondrion
centriole
vesicle
cytosol
chromatin
nuclear envelope
nuclear pore
nucleolus
nucleus
Golgi complex
lysosome
flagellum
plasma membrane
rough endoplasmic reticulum
ribosomes
smooth endoplasmic reticulum

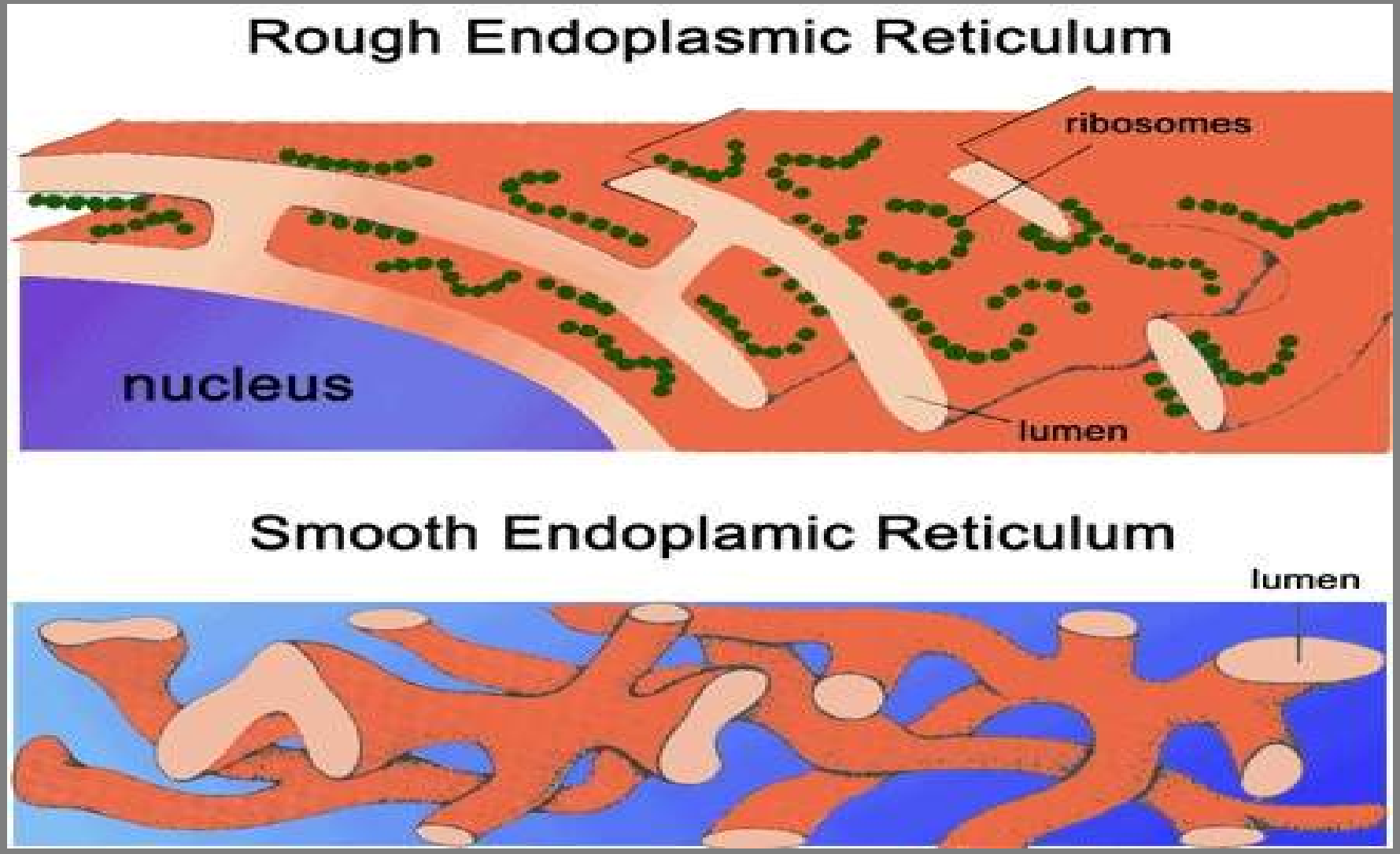
The Cell


Ribosomes
Centrioles
Centrosome
Smooth endoplastic reticulum
Mitochondria
Smooth endoplastic reticulum
Cilia
Mitochondrion
Lysosome
Rough endoplastic reticulum
Free ribosomes
Peroxisome
Golgi apparatus
Cytoskeleton
Microvilli
Vesicle
Intermediate filament
Nuclear envelope
Nucleus
Nucleolus
Vault
Microtubule
Microfilament

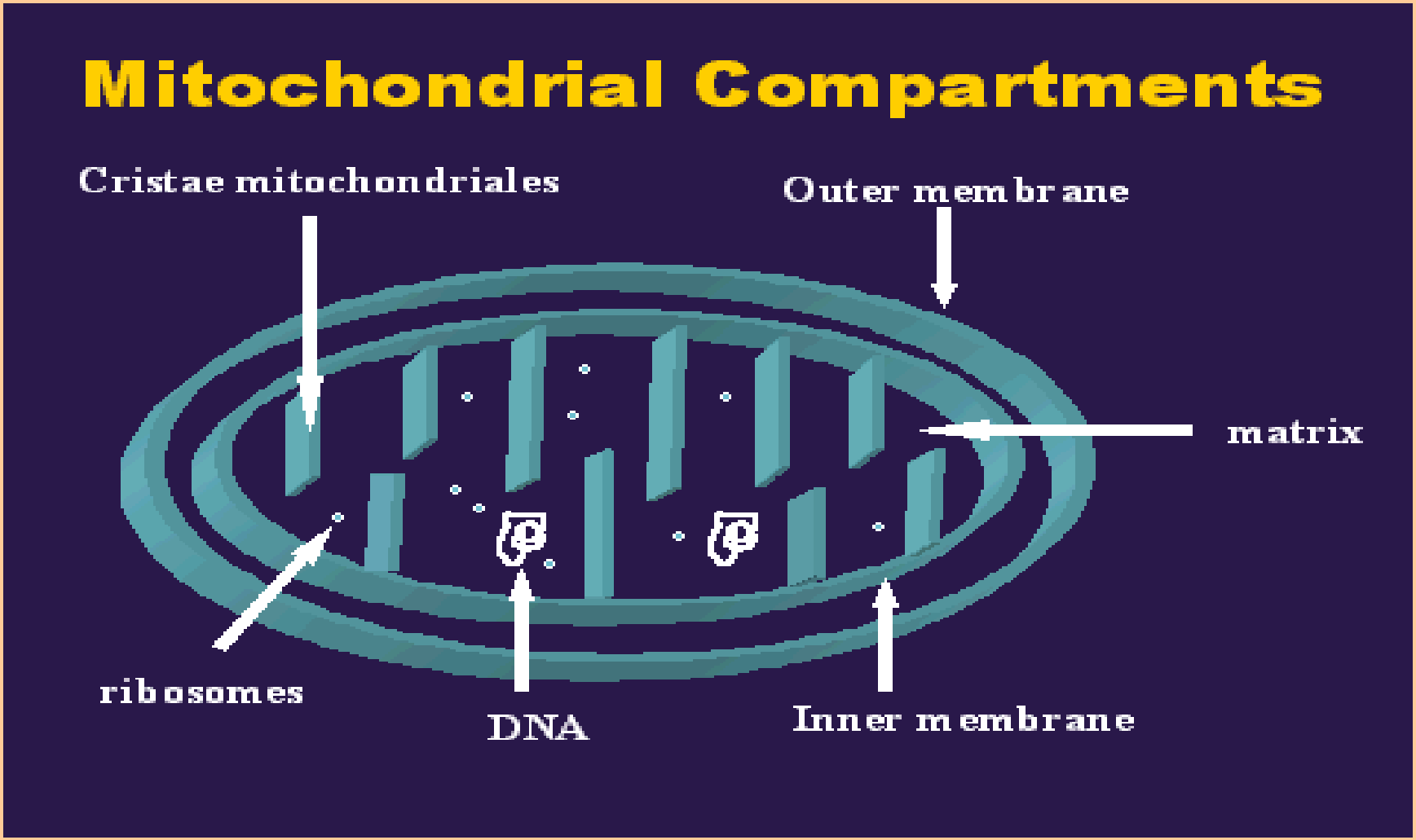
Step 1. Three Na+ ions bind to cytoplasmic high-affinity binding sites.



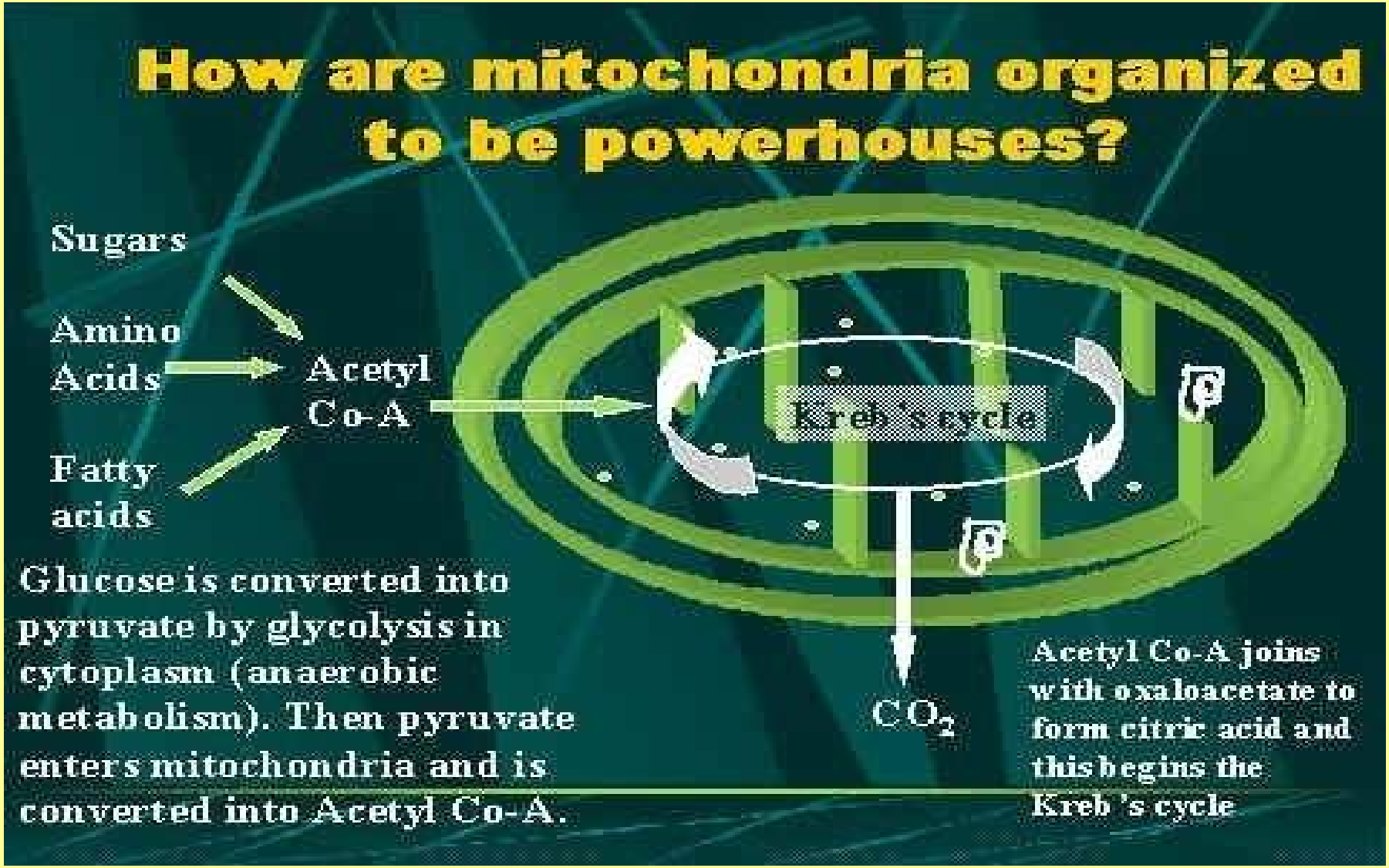
Endoplasmic reticulum



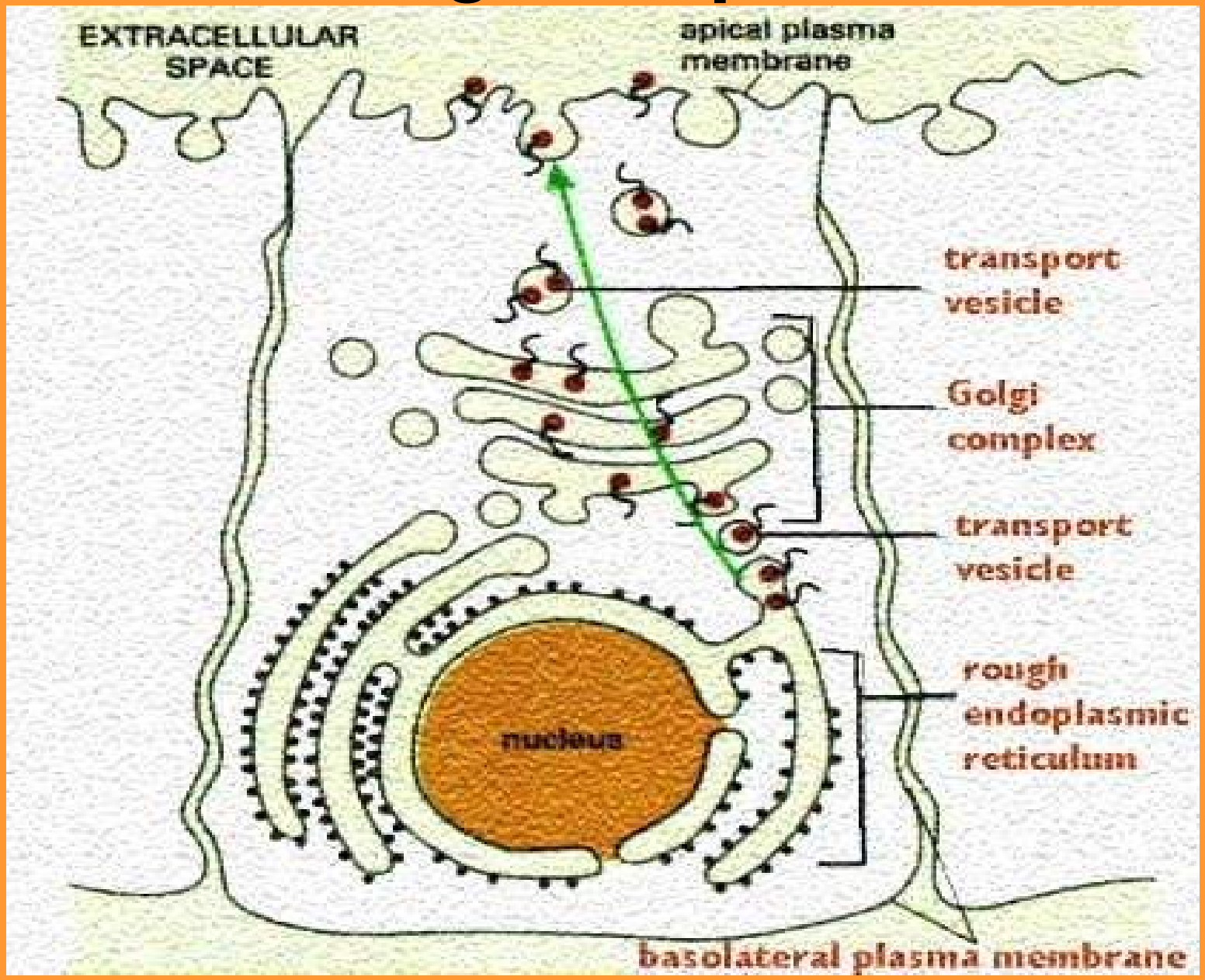
Mitochondria

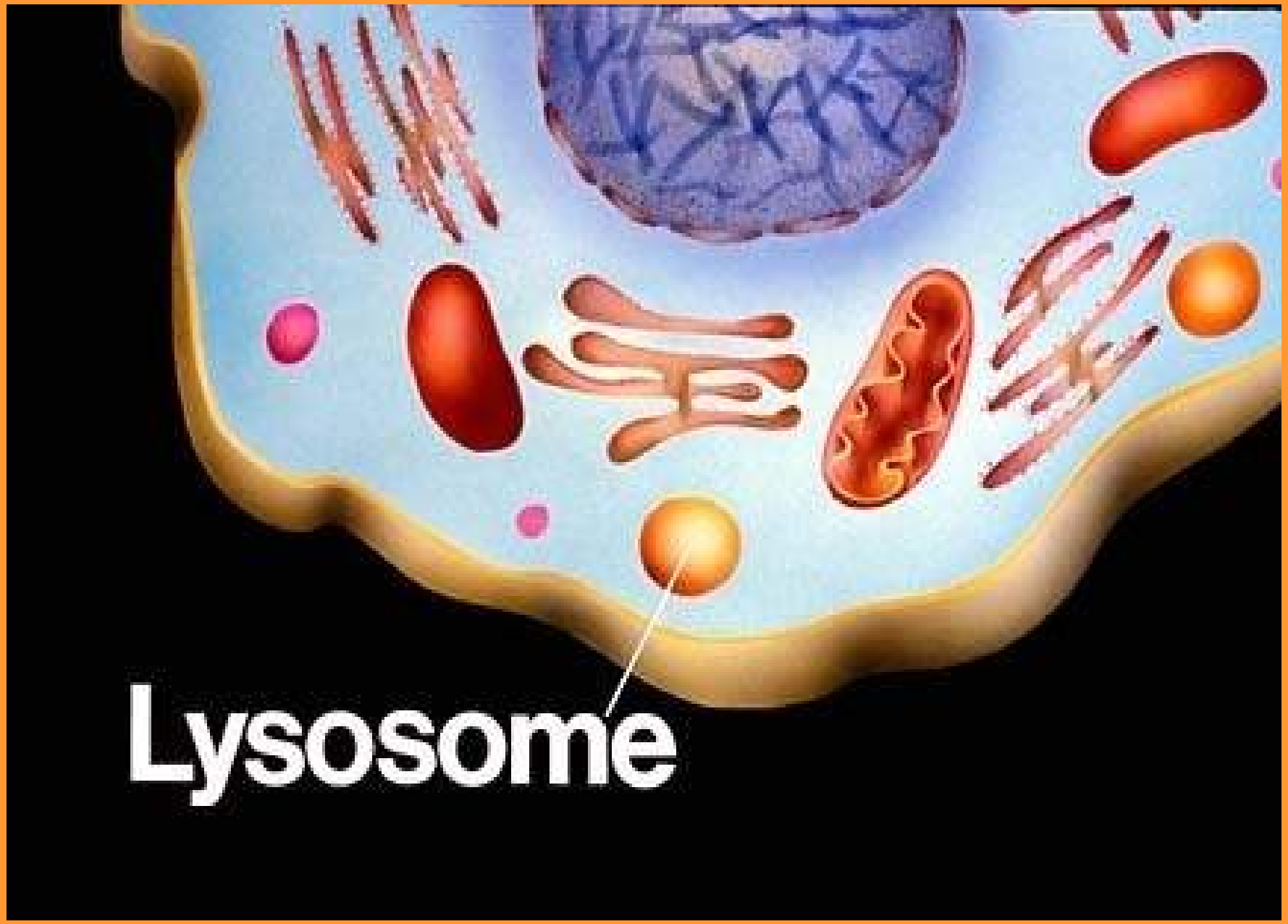


Mitochondria

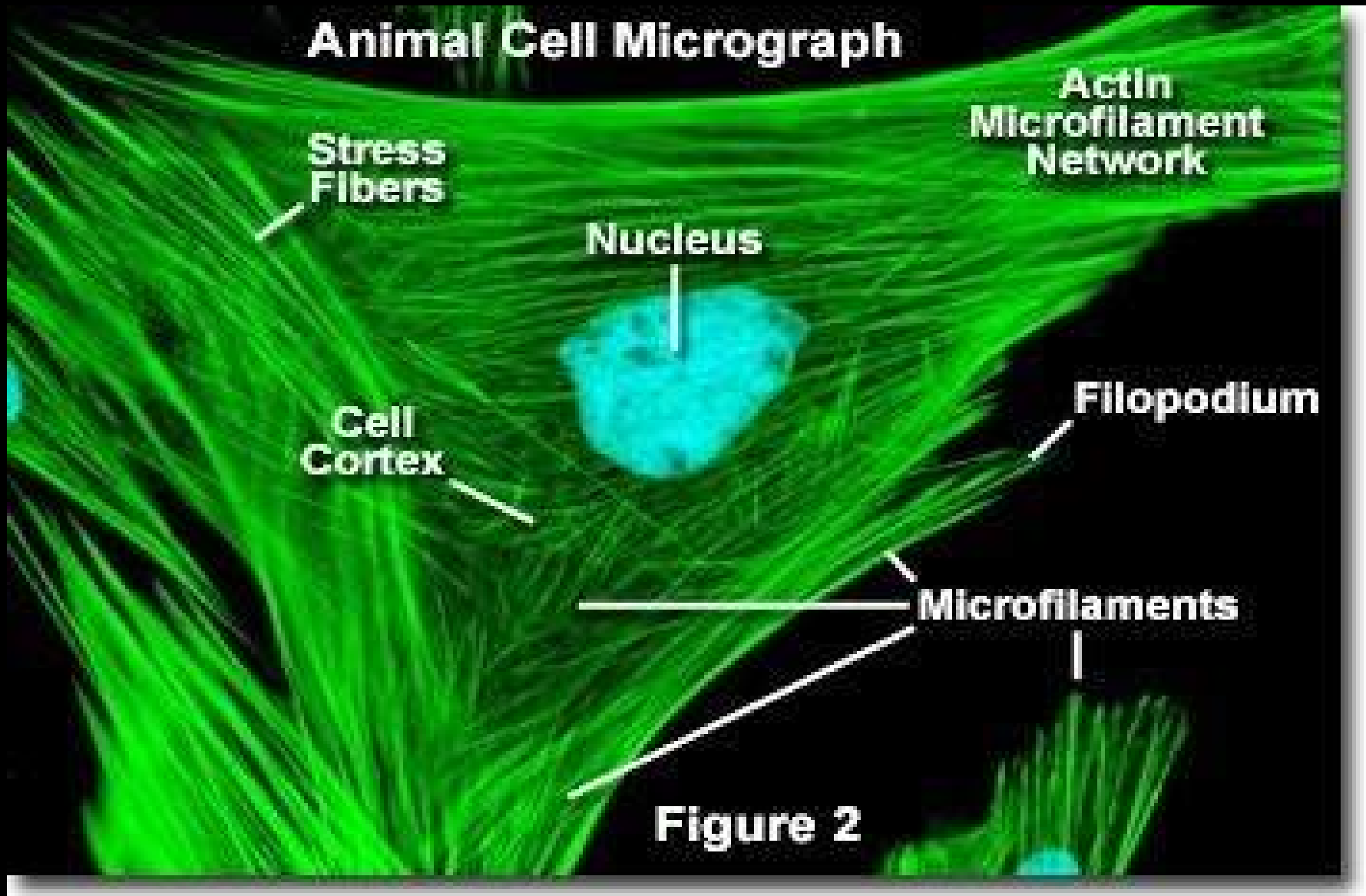


Golgi complex




Lysosome

Microfilaments



Cell Organelles & Their Functions

Organelle	Function	Biochemical Systems
Nucleus	Manufacture of nucleic acids	Nucleic acids, lipids, proteins
Nucleoli	Manufacture of RNA and proteins	RNA, proteins
Ribosomes	Manufacture of proteins	RNA, proteins
Endoplasmic reticulum	Manufacture of proteins	RNA, proteins
Lysosomes	Defense	Proteins (enzymes)
Membranes	Regulatory	Lipids, proteins, carbohydrates
Mitochondria	Oxidative reactions, electron transport	Nucleic acids, Coenzymes, ions in organic-rich systems
Golgi net	Packaging, transport, CHO metabolism	Proteins, lipids, carbohydrates